

iSE

Internal Kickoff & Team Matching

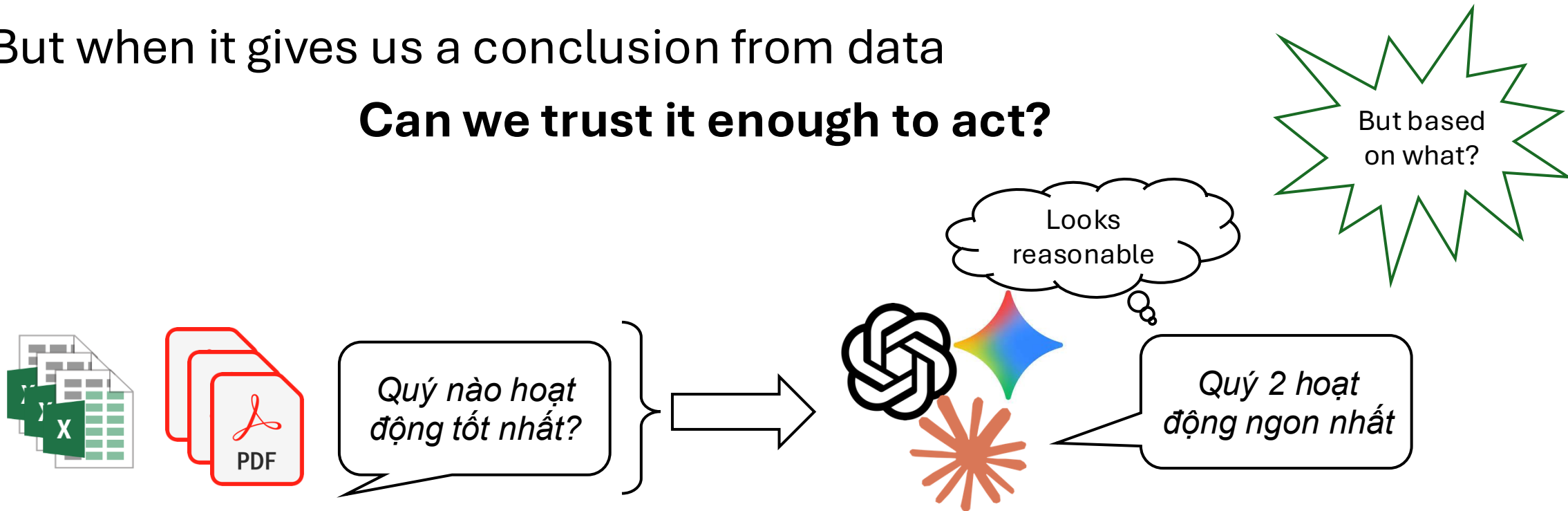
AXIOM

From Answering to Accountable Data Intelligence

Illusion of Intelligence

- AI (LLMs) can read documents, write code, analyze tables, and generate charts.
- But when it gives us a conclusion from data

Can we trust it enough to act?



When Intelligence Looks Right but Goes Wrong

What AI appears to do well

- Fluent answer
- Code generation
- Chart generation
- Confident explanation

What may silently go wrong

- Misunderstood KPI
- Wrong table join
- Temporal leakage
- Dirty data
- Wrong aggregation level

The most dangerous failure is the one that still looks plausible!

An iceberg floating in the ocean. The tip of the iceberg is above the water line, while the much larger base is submerged. The water is a deep blue, and the sky is a lighter blue with some wispy clouds.

Fluent answer
Code generation
Chart generation
Confident explanation

Misunderstood KPI
Wrong table join
Temporal leakage
Dirty data
Wrong aggregation level

A Simple Question. Many Hidden Assumptions.

“Hoạt động tốt” nghĩa là doanh thu, lợi nhuận, tăng trưởng hay retention?

Theo ngày ghi nhận đơn, ngày thanh toán hay ngày giao hàng?

Quý nào hoạt động tốt nhất?

Tính theo chi nhánh, sản phẩm hay toàn hệ thống?

Có so sánh cùng kỳ không?

Có loại bỏ đơn hoàn/hủy không?



Real Data Is Not Analysis-Ready

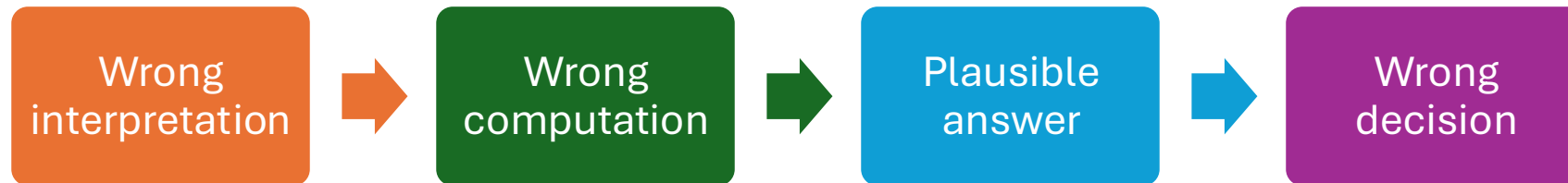
- Data is scattered, inconsistent, incomplete, and semantically ambiguous.
 - Complex spreadsheets
 - Documents/forms/images
 - Disconnected sources
 - Implicit business rules
 - Missing/noisy/duplicated data



The data is messy before the intelligence even begins.

Fluent Answers Are Not Enough

A **fluent answer** is not necessarily a correct answer
A **correct-looking analysis** can still lead to a wrong decision

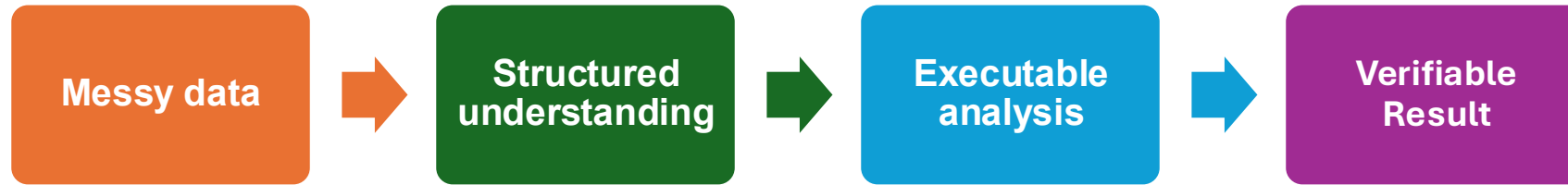


Existing Approaches

Approach	Strength	Missing
<i>Text-to-answer</i>	Fast, flexible	Hard to inspect/reproduce
<i>Text-to-code-to-answer</i>	Executable, debuggable	Execution != correctness
<i>AutoML/Data tools</i>	Strong for prepared data	Often assume clean schema

A program can execute perfectly while answering the wrong question

The Missing Layer

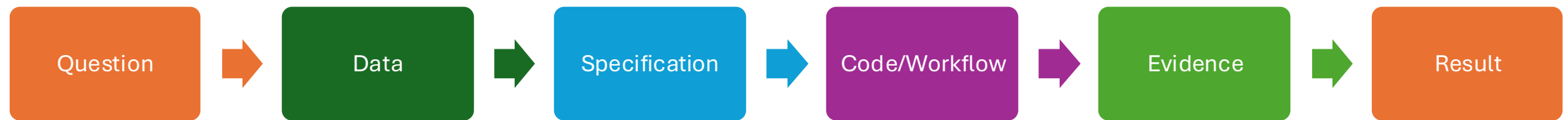


- **Messy Data:** heterogeneous, incomplete, ambiguous
- **Structured Understanding:** entities, schema, constraints
- **Executable Analysis:** code, workflow, computation
- **Verifiable Result:** evidence, reproducibility, auditability

The gap is not just a stronger model!
The gap is an accountable data intelligence layer

AXIOM: Transparency by Design

Every important data-driven conclusion should be traceable



AXIOM does **not replace** human decision-making.
It makes AI-assisted decisions **inspectable and trustworthy!**

Three Principles

Accountable Data Intelligence



Data-centric



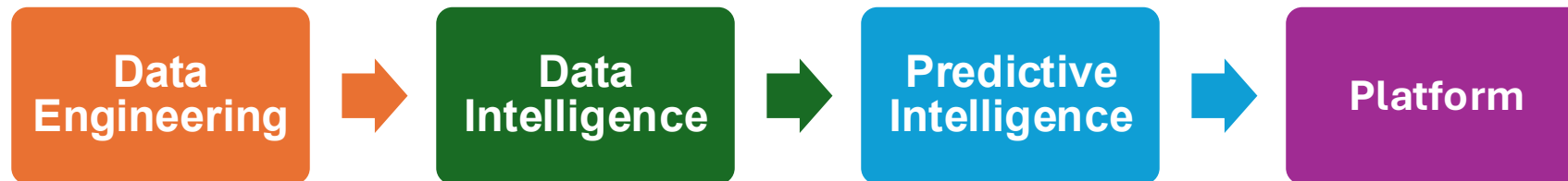
Specification-first



Code-driven

AXIOM Research Program

- **Data Engineering Research:** semi-structured/messy data foundations
- **Data Engineering R&D:** operational data pipelines
- **Data Intelligence Research:** code-driven analysis
- **Data Intelligence R&D:** user-facing analytical workflows
- **Platform AXIOM:** integration, execution, evaluation

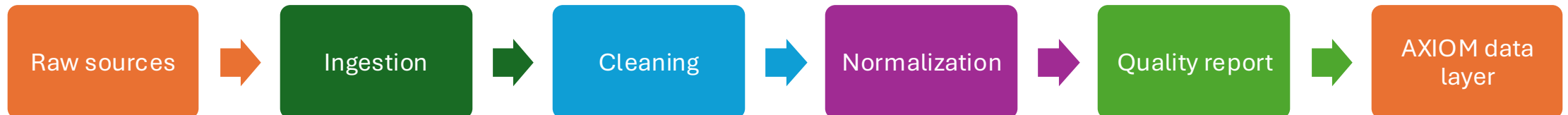


Project 1: Data Engineering Research

- Active Project: *Semi-structured Tables*
- Members: *Minh Quân, Lê Anh Duy, Đậu Đức Hiếu*
- Objective: *Turn complex tables into semantically structured, analysis-ready data.*
- RQ: *How can we recover entities, relations, headers, and constraints from messy tables?*

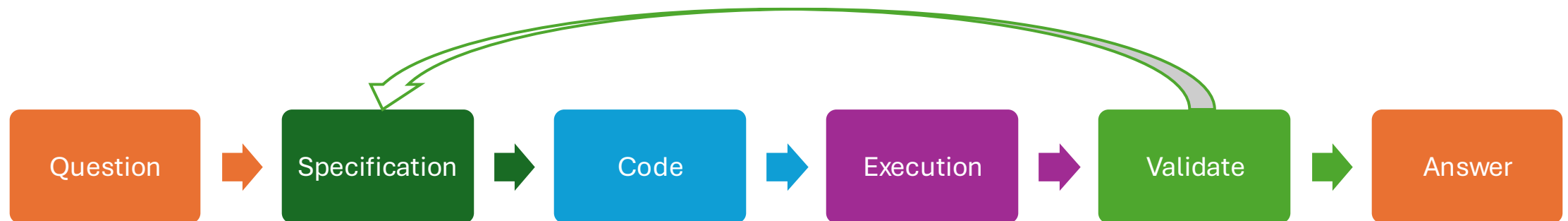
Project 2: Data Engineering R&D

- Members: *Nam, Đăng + 2 SV mới*
- Objective: *Build robust data ingestion, cleaning, normalization, and integration pipelines for real AXIOM use cases.*
- Focus:
 - data ingestion;
 - normalization;
 - quality checks;
 - integration with AXIOM platform;



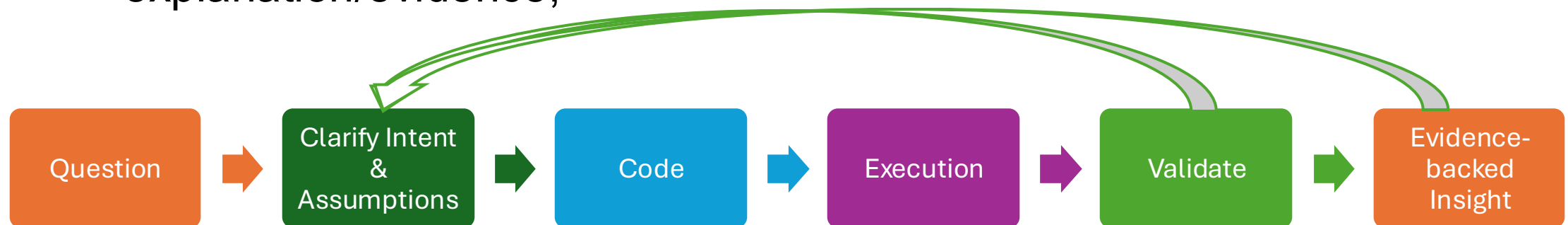
Project 3: Data Intelligence Research

- Active Project: *Code-Driven Data Analysis*
- Leader: *Duy Quân, Trần Anh Duy, Thảo Linh*
- Objective: *Generate not just answers, but verifiable analytical workflows.*
- RQ: *How can AI translate analytical intent into executable, constraint-aware, and checkable code?*



Project 4: Data Intelligence R&D

- Members: *Tuấn Anh, Đạt, Cường, Sơn + 2 SV mới*
- Objective: *Turn code-driven data intelligence methods into real analytical workflows that users can inspect, refine, and trust.*
- Focus:
 - analytical workflows;
 - enterprise scenarios;
 - explanation/evidence;



Project 5: Platform AXIOM

- Members: *Hiếu + Thủy, Vinh, Vũ, Việt, Giáp, Lưu + 1 SV mới*
- Objective: *Integrate data, workflow generation, execution, validation, and evidence synthesis into a unified platform.*
- Focus:
 - architecture;
 - execution environment;
 - data management & pipeline/workflow;
 - others

Beyond AXIOM

Project 6: Chỉnh lý tự động

- Thành viên: *Hiếu, Vinh, Vũ, Thịnh, Hưng, Bình + 2 SV mới*
- Mục tiêu: *Tự động đọc lượng lớn văn bản, trích xuất nội dung, phân tích, phân loại tài liệu tự động*
- Tập trung:
 - Dữ liệu thật
 - Bài toán siêu lớn
 - OCR chuẩn hiệu năng cao
 - Tác động xã hội

Project 7+8: Số hoá + SiFLEX

- Thành viên: *Thuỷ, Duy Quân, Minh Quân, Lê Anh Duy, Tuấn Anh + 1 SV mới*
- Mục tiêu: *Xây dựng pipeline số hóa và khai thác dữ liệu/tài liệu theo hướng có cấu trúc, có truy vết và có khả năng tích hợp vào các hệ thống phân tích.*
- Tập trung:
 - Dữ liệu thật
 - Bài toán siêu lớn
 - OCR chuẩn hiệu năng cao
 - Tác động xã hội

Project 9: Fault localization (Denso)

- Thành viên: *Hà Linh, Đức, Hồng Anh*
- Mục tiêu: *Hỗ trợ định vị nguyên nhân lỗi trong hệ thống phần mềm/kỹ thuật bằng cách kết hợp tín hiệu thực thi, log, mã nguồn và tri thức miền.*
- Tập trung:
 - bug/fault localization;
 - log and trace analysis;
 - ranking suspicious components;
 - explainable diagnostic evidence;
 - industrial evaluation with Denso scenarios.

Project 10: MAS failure detection

- Thành viên: *Hiếu, Vinh, Thịnh, Hưng, Bình*
- Mục tiêu: *Phát hiện, phân loại và giải thích các lỗi phát sinh trong hệ thống nhiều tác tử, nơi lỗi có thể đến từ phối hợp sai, vai trò mơ hồ, hoặc lan truyền thông tin không đúng.*
- Focus:
 - failure taxonomy;
 - interaction trace analysis;
 - anomaly detection in agent collaboration;
 - diagnosis and recovery suggestions;
 - benchmarks for MAS reliability.

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